

Inventory Management System Project Report

Python, Java, and Data Analytics Implementation

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WEEK 3: May 18, 2026 – May 24, 2026

Project Overview

The Inventory Management System project focuses on designing and implementing an efficient software solution to manage stock levels, product tracking, forecasting, and reporting. The system integrates analytics tools to improve inventory decisions and operational efficiency.

Objectives

- Develop REST API controllers for product, supplier, transaction, and authentication modules.
- Implement user authentication and login functionality for system access control.
- Build a basic frontend dashboard for inventory management operations.
- Enhance analytics and reporting features for inventory monitoring and forecasting.

Roles and Responsibilities

- Developed Spring Boot REST controllers for inventory and supplier management APIs.
- Implemented user authentication and login validation using Java and SQLite.
- Created a frontend dashboard starter interface using HTML, CSS, and JavaScript.
- Performed API testing using Postman and updated project documentation and reporting.

Tools and Technologies Used

Python, Java, Spring Boot, SQLite, SQL, Pandas, Matplotlib, GitHub, VS Code, Postman

Key Outcomes

- Successfully implemented backend REST APIs and authentication modules.
- Developed a basic dashboard interface and integrated API testing workflows.
- Improved understanding of API development, frontend integration, and system architecture.

Conclusion

Week 3 focused on expanding the Inventory Management System with REST APIs, authentication, frontend integration, and testing workflows. The completed tasks significantly improved the system structure and prepared the project for advanced feature development and deployment.

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